OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

15.4 Stations

Rail-bound vehicles like trams need stations for people to get on and off. Each station shall have at least one platform, which may be further divided into segments. The platforms determine the physical extent of a station.

The `<station>` element may also be used for bus stations.

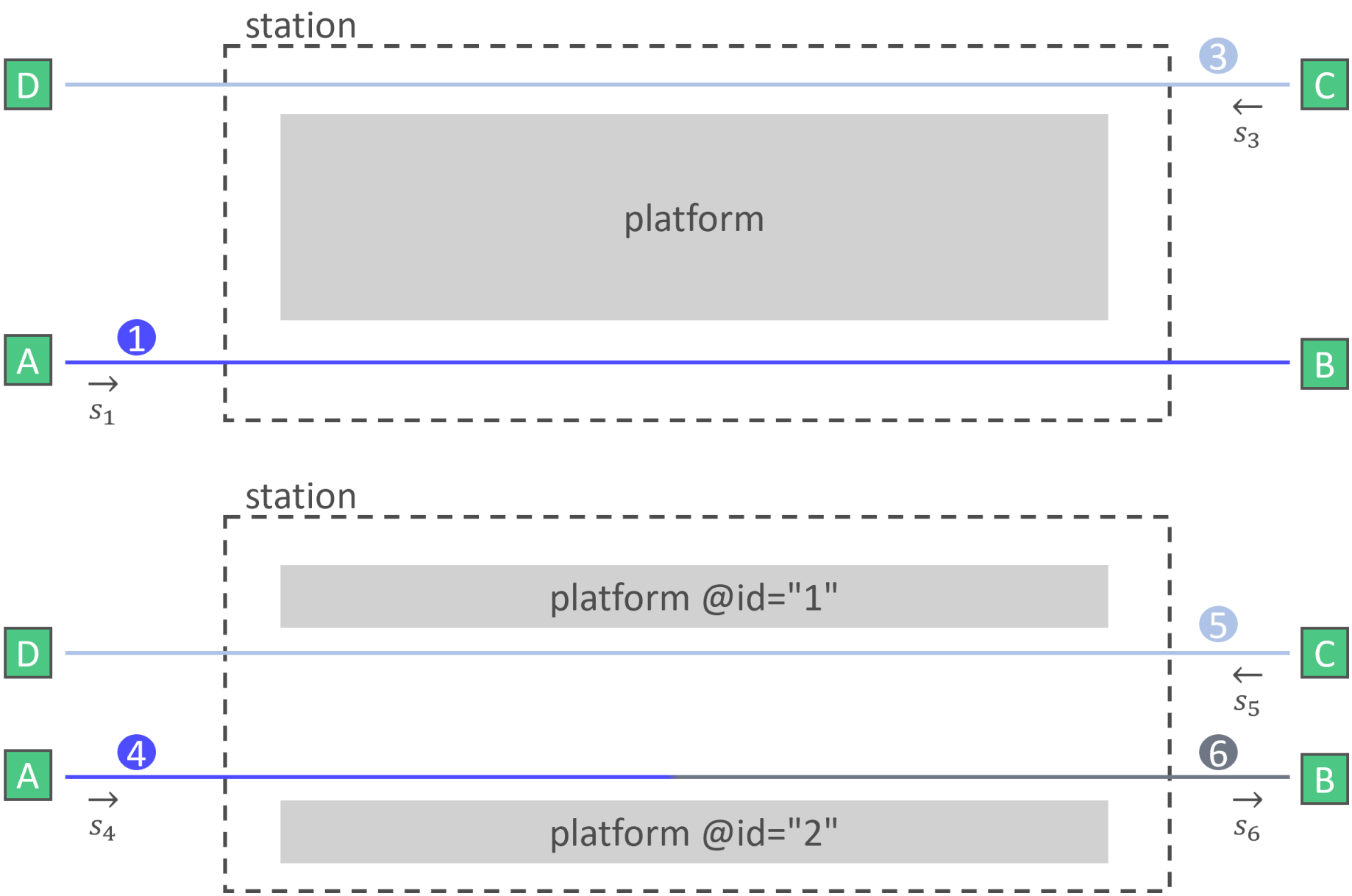

Figure 142. Railroad stations

Figure 142 shows two scenarios for stations:

- In the first scenario, one platform is referenced by the roads `1` and `3`, running in different driving directions. The platform consists of one segment only.
- In the second scenario, platform `1` is referenced by road `5` only. Platform `2` is referenced by road `4` and `6`. Platform `2` is split into two segments.

Elements in UML model

`<station>` element

In ASAM OpenDRIVE, stations are represented by the `<station>` element within the `<OpenDRIVE>` element.

```
UML class: t_station
XML tag:   <station> (Multiplicity: 0..*)
```

Stations are places on the rail network where passengers enter and leave rail-bound vehicles at platforms.

May refer to multiple tracks and is therefore defined on the same level as junctions.

Table 146. Attributes of the <station> element

Name	Type	Use	Description
`id`	string	required	Unique ID within database
`name`	string	required	Unique name of the station
`type`	e_station_type	optional	Type of station. Free text, depending on the application. e.g.: small, medium, large

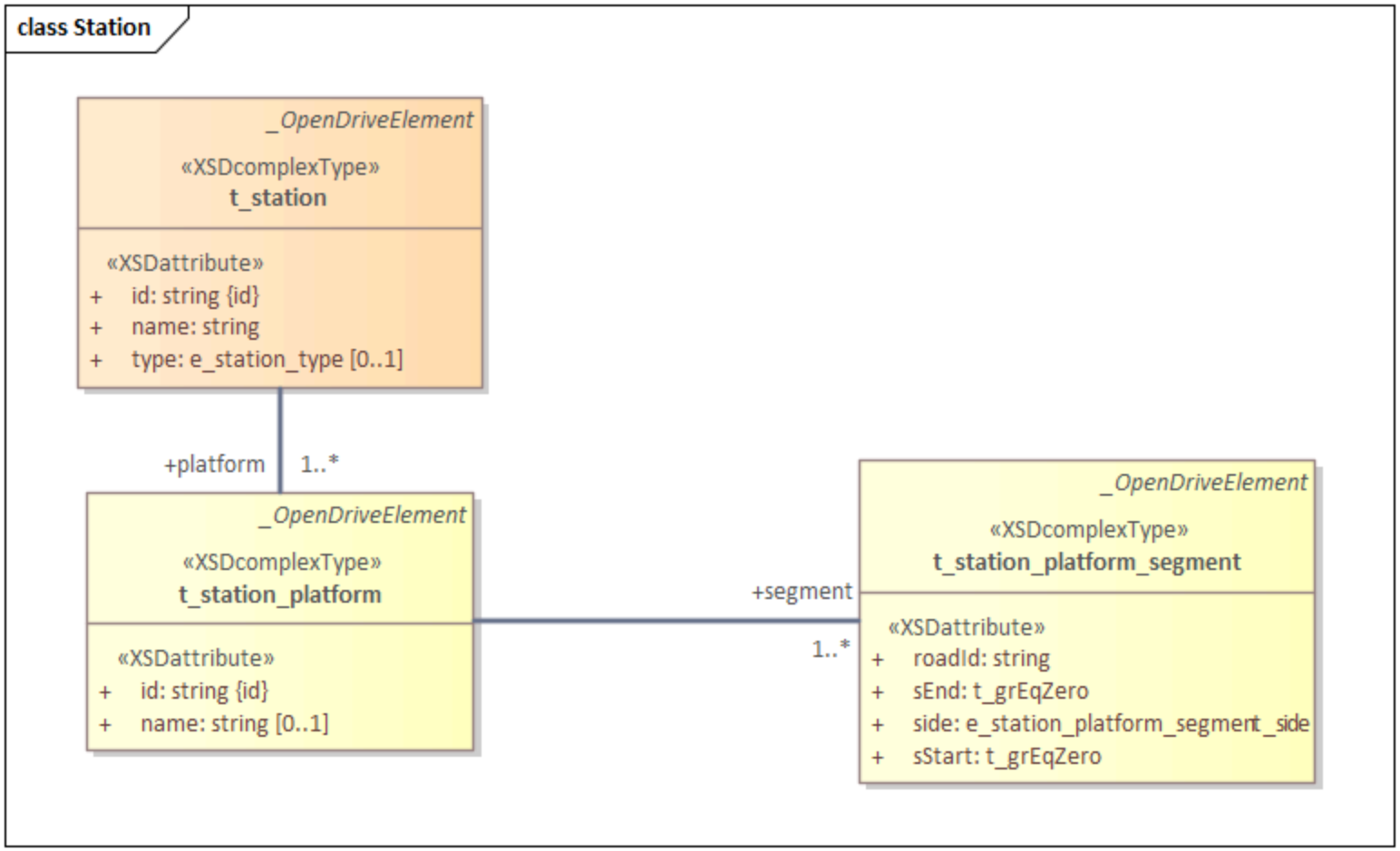

Figure 143. UML class diagram of the Station class

Figure 143 shows the UML class diagram of the ASAM OpenDRIVE Station class.

XML example

- Ex_Railway-Station.xodr

Rules

The following rules apply to stations:

- asam.net:xodr:1.7.0:road.railroad.stations.one_platform_per_station: A `<station>` element shall be followed by at least one `<platform>` element.
- The type of the station may be further specialized by the @type attribute. The values are stored in the used application.

Related topics

- Section 15.1, "Introduction to railroads"
- Section 15.4.1, “Platforms”
- Section 15.4.2, “Segments”

15.4.1 Platforms

A station shall contain at least one platform. A platform shall be referenced by one or more railroad tracks. See picture in Figure 142.

Elements in UML model

`<platform>` element

In ASAM OpenDRIVE, platforms are represented by the `<platform>` element within the `<station>` element.

```
UML class: t_station_platform
XML tag:   <platform> (Multiplicity: 1..*)
```

Platforms are essential parts of stations for passengers to enter and leave rail-bound vehicles. One or more railroad tracks reference one platform.

Table 147. Attributes of the <platform> element

Name	Type	Use	Description
`id`	string	required	Unique ID within database

Name	Type	Use	Description
`name`	string	optional	Name of the platform. May be chosen freely.

Rules

The following rules apply to platforms:

- asam.net:xodr:1.7.0:road.railroad.platforms.min_amount: There shall be at least one platform per station.
- asam.net:xodr:1.7.0:road.railroad.platforms.min_segments: A platform shall contain at least one segment.

Related topics

- Section 15.4.2, “Segments”

15.4.2 Segments

Platforms may be further divided into segments. This is useful if a bi-directional railroad track runs along the same platform. A platform shall contain at least one segment.

Elements in UML model

`<segment>` element

In ASAM OpenDRIVE, segments are represented by the `<segment>` element within the `<platform>` element.

```
UML class: t_station_platform_segment
XML tag:   <segment> (Multiplicity: 1..*)
```

Segments are parts of platforms.

Each `<platform>` element is valid on one or more track segments. The `<segment>` element must be specified.

Table 148. Attributes of the element

Name	Type	Use	Unit	Description
`roadId`	string	required		Unique ID of the `<road>` element (track) that accompanies the platform
`sEnd`	t_grEqZero	required	m	Maximum s-coordiante on `<road>` element that has an adjacent platform
`sStart`	t_grEqZero	required	m	Minimum s-coordinate on `<road>` element that has an adjacent platform
`side`	e_station_platform_segment_side	required		Side of track on which the platform is situated when going from sStart to sEnd

Rules

The following rules apply to segments:

- asam.net:xodr:1.7.0:road.railroad.segment.segmenta_per_platform_min_amount: There shall be at least one segment per platform.

Related topics

- Section 15.4.1, “Platforms”